A **coefficient** is a **number that multiplies a variable** in a mathematical equation or a machine learning model.

In Mathematics

In the equation:

$5x+3y-8=0$

- **5** is the coefficient of **x**
- **3** is the coefficient of **y**
- **-8** is a constant (not a coefficient)

Another example:

$-7a+4b=20$

- Coefficient of **a** = **-7**
- Coefficient of **b** = **4**

Coefficient in Linear Regression

In machine learning, a coefficient tells you **how much the target variable changes when a feature changes by one unit**, assuming all other features remain constant.

The linear regression equation is:

$Y=\beta0+\beta1X1+\beta2X2+\cdots+\beta nXn$

Where:

- Y = Predicted value
- $\beta0$ = Intercept
- $\beta1,\beta2,\ldots$ = Coefficients
- X1,X2,… = Features

Example

Suppose the model is:

$Salary=25000+5000\times Experience$

Here:

- Intercept = **25000**
- Coefficient of Experience = **5000**

Interpretation:

- Every additional **1 year of experience** increases the predicted salary by **₹5,000**.

Positive and Negative Coefficients

Positive Coefficient

$Y=10+4X$

Coefficient = **+4**

As **X increases**, **Y increases**.

Example:

X	Y
1	14
2	18
3	22

Negative Coefficient

$Y=100-5X$

Coefficient = **-5**

As **X increases**, **Y decreases**.

Example:

X	Y
1	95
2	90
3	85

Coefficients in Python (Scikit-learn)

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
```

Example output:

```
Intercept: 15000
Coefficients: [4500, -200, 800]
```

Suppose your features are:

```
Experience
Age
Education
```

Then:

Feature	Coefficient	Meaning
Experience	4500	Salary increases by ₹4,500 for each additional year of experience.
Age	-200	Salary decreases by ₹200 for each additional year of age (holding other features constant).
Education	800	Salary increases by ₹800 for each additional education level.